

Question

Iodometry is an important experimental method in redox titration. The laboratory attempted to determine the equilibrium constant of the reaction $I_2 + I^- = I_3^-$ using iodometry, with the procedure as follows:

Weigh three portions of 0.5 g of finely ground iodine into three numbered ground-glass conical flasks labeled 1, 2, and 3. Add 80.0 mL of KI solution with a concentration of $c_0(\text{mol} \cdot \text{L}^{-1})$ to flask 1; add 80.0 mL of KI solution with a concentration of $2c_0(\text{mol} \cdot \text{L}^{-1})$ to flask 2; and add 200.0 mL of deionized water to flask 3. Add magnetic stir bars, stopper the flasks tightly, and stir the reaction at room temperature for 30 minutes. After the reaction, take the solutions from the three flasks and centrifuge to obtain the supernatant. Pipette V_s (mL) of the centrifuged solutions from flasks 1 and 2, dilute each with water to 50 mL, and titrate with a standard solution of $\text{Na}_2\text{S}_2\text{O}_3$ at a concentration of c ($\text{mol} \cdot \text{L}^{-1}$). The volumes of $\text{Na}_2\text{S}_2\text{O}_3$ standard solution consumed are V_1 (mL) and V_2 (mL), respectively. Then, pipette $5V_s$ (mL) of the centrifuged solution from flask 3 and titrate it with the same $\text{Na}_2\text{S}_2\text{O}_3$ standard solution, consuming V_0 (mL) of the standard solution.

Using the given parameters c, c_0, V_0, V_1, V_s , the equilibrium constant K of the reaction can be calculated.

Which of the following operational errors would lead to an overestimation of the equilibrium constant K ?

1. The masses of the three portions of iodine are slightly different.
2. After stirring, the solution in flask 3 becomes clear and transparent.
3. When pipetting the solution from flask 1, solid settled at the bottom is accidentally drawn into the pipette.
4. While preparing the KI solution, a small amount of solid KI is accidentally spilled after weighing.
5. The $\text{Na}_2\text{S}_2\text{O}_3$ standard solution was left for too long after standardization.

A. 1, 2, 3

B. 3, 5

C. 1, 4

D. 2, 3, 4

E. 2, 3, 5

F. 2, 3

G. 2, 3, 4, 5

H. 3, 4

Answer

Correct Answer: **E**

Detailed Explanation

The data from Erlenmeyer flasks 1 and 3 can be combined to calculate the equilibrium constant K . Flask 3 provides the saturated concentration of I_2 , and together with the data from flask 1, the equilibrium constant for the reaction $I_2 + I^- = I_3^-$ can be obtained. The derivation yields $K = (5V_1 - V_0) * V_s / [(c_0V_s - 0.5cV_1 + 0.1cV_0) * V_0]$.

Analysis of the impact of operational errors on K :

1. Since iodine is in excess in the experiment, its mass has no effect on K .

CHECKPOINT

1 PTS

Iodine is in excess, so its mass has no effect on K .

2. If flask 3 becomes clear and transparent after stirring, it indicates insufficient iodine was added, resulting in a measured saturated iodine concentration that is too low, leading to an overestimated equilibrium constant K .

CHECKPOINT

1 PTS

Error 2 causes the measured saturated iodine concentration to be too low, resulting in an overestimated K .

3. If solid settled at the bottom is accidentally drawn into the pipette when sampling solution from flask 1, the measured I_3^- concentration will be too high, leading to an overestimated equilibrium constant K .

CHECKPOINT

1 PTS

Error 3 causes the measured I_3^- concentration to be too high, resulting in an overestimated K .

4. If a small amount of KI solid is accidentally spilled after weighing during the preparation of the KI solution, the actual KI concentration will be less than c_0 , meaning the c_0 used in calculations is overestimated. Based on the expression for K , an overestimated c_0 leads to an underestimated K .

CHECKPOINT

1 PTS

Error 4 causes c_0 to be overestimated, resulting in an underestimated K .

5. If the standardized $\text{Na}_2\text{S}_2\text{O}_3$ solution is left standing for too long after calibration, its concentration will decrease, meaning the c used in calculations is overestimated. Based on the expression for K , an overestimated c leads to an overestimated K .

CHECKPOINT

1 PTS

Error 5 causes c to be overestimated, resulting in an overestimated K .

Therefore, the operations that lead to an overestimated K are 2, 3, and 5.